

Olympiad Test Paper — Science (Class 7) — Paper 3

Chapter 14: Electric Current and Its Effect

Paper 3 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	14 — Electric Current and Its Effect
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Paper 3 traps: confusing the heating effect with the magnetic effect; assuming more cells or a thicker fuse always helps; thinking an electromagnet keeps its pull after the current stops; believing a fuse 'conducts away' surplus current rather than melting; assuming any complete-looking loop must light a bulb even when it shorts past the bulb or includes an insulator/reversed cells; thinking copper heats more than nichrome; thinking a compass deflects without current; mistaking the cause and effect of Oersted's needle swing. Do not write on this paper — mark answers on your answer sheet.

1. On a circuit-diagram cell symbol the longer line is drawn thin and the shorter line is drawn thick. A student labels the thick short line as the positive terminal. Why is this wrong?

- (A) Both lines stand for connecting wires, so neither is a terminal at all.
- (B) By convention the longer thin line is the positive (+) terminal and the shorter thick line is the negative (-) terminal, so the student has them swapped.
- (C) The thickness shows the strength of the cell, not which terminal is positive.
- (D) The longer line is always drawn on the right and so must be negative.

2. Three identical 2 V cells are to be joined into a battery so their voltages add to 6 V. Which joining rule achieves this?

- (A) Join every positive terminal to a common point and every negative terminal to another common point.
- (B) Join positive terminal to positive terminal and negative to negative along the row.

(C) Join them in a row so that the positive terminal of each cell connects to the negative terminal of the next.

(D) Place the three cells in the holder with small air gaps so they do not touch.

3. A torch needs two cells. By mistake the second cell is inserted facing the first positive-to-positive. The bulb does not glow. The best explanation is that:

(A) Each cell on its own is too weak, and facing them this way makes them even weaker than zero.

(B) The two cells push current in opposite directions, so their effects cancel and almost no net current flows.

(C) Facing positive-to-positive doubles the voltage, which instantly burns out the filament.

(D) Two cells can never light a bulb; a torch must use exactly one cell.

4. A bulb glows on one cell. A second identical cell is added in series the right way (+ to -). The current increases and the bulb brightens. The student adds a third cell the same way. The most likely result is that the bulb:

(A) Goes dimmer, because the three cells must share the available current between them.

(B) Glows even brighter, and with too many cells the extra current may eventually fuse the filament.

(C) Stays exactly as bright as with two cells, because brightness has a fixed maximum from the first cell.

(D) Goes out, because an odd number of cells cannot complete a circuit.

5. In a circuit diagram the cell is drawn with its long thin line on the left. In which direction is conventional current taken to flow through the external wires?

(A) Out from the long thin line (positive terminal), round the external circuit, and back into the short thick line (negative terminal).

(B) Out from the short thick line (negative terminal) and back into the long thin line.

(C) From the middle of the cell symbol outward in both directions at once.

(D) Only from the bulb toward the switch, never back to the cell.

6. A diagram shows a small circle with a cross inside it placed in a loop with a cell and a closed switch. What does this symbol represent, and what should happen?

(A) It is a cell, so it adds extra voltage to the circuit.

(B) It is an open switch, so the circuit is broken and nothing works.

(C) It is a bulb, and because the loop is closed it should glow.

(D) It is a fuse, so it is there only to break the circuit when current is high.

7. A learner draws a switch as a small gap in the wire with a hinged arm lying flat, touching both contacts. What state is the switch in, and what is the effect on a bulb in that loop?

- (A) The switch is open, so the loop is broken and the bulb stays dark.
- (B) The arm lying flat means the switch has melted like a fuse, so it must be replaced.
- (C) A flat arm means the cell is being recharged, so the bulb is dim.
- (D) The switch is closed, the loop is complete, so the bulb can glow.

8. A circuit is wired: cell → wire → bulb → wire → open switch → back to cell. A student insists the bulb will glow because 'there is a bulb and a cell present.' Why is the student wrong?

- (A) The bulb glows only if the cell is also being charged at the same time.
- (B) An open switch leaves a gap in the loop, so the circuit is not complete and no current can flow through the bulb.
- (C) The bulb would glow, but the wires absorb all the current before it reaches the bulb.
- (D) The bulb cannot glow because a single cell never has enough push for any bulb.

9. A bulb is connected to a cell, but a short, thick, low-resistance wire is also joined directly across the bulb's two ends (a short across the bulb). What is most likely observed?

- (A) The bulb glows even brighter, because the extra wire feeds it more current.
- (B) Nothing changes, because an extra wire can never affect a working bulb.
- (C) The cell instantly stops working forever, even after the extra wire is removed.
- (D) The bulb stays dark because almost all the current takes the easy short-circuit path around it instead of through the filament.

10. In a single-loop circuit with a fresh cell, a closed switch and a good bulb, the bulb still does not light. One connecting wire has a hairline break inside its insulation. What kind of circuit is this, and why no glow?

- (A) It is a closed circuit that should be glowing, so the bulb itself must simply be the wrong colour.
- (B) It is a short circuit, which makes the bulb shine more brightly than normal.
- (C) It is an open circuit; the hidden break leaves a gap, so the loop is incomplete and no current flows.
- (D) It has become a two-cell battery, so the bulb has too much voltage and is off.

11. A loop contains a cell, a bulb, a closed switch and — in series in the same loop — a short strip of dry wood. The bulb does not glow. The wood is responsible because it:

- (A) Is such a good conductor that it carries all the current away from the bulb.
- (B) Is a low-melting metal that melts and breaks the loop like a fuse.
- (C) Is magnetic, and magnetism stops electric current from flowing.
- (D) Is an insulator, so it blocks the loop and prevents current from completing its path.

12. A student replaces the metal connecting wire in a working bulb circuit with a strip of dry rubber, expecting the bulb to glow more safely. Instead the bulb goes dark. Why?

- (A) Rubber is a better conductor than metal, so it short-circuits the bulb and hides its glow.
- (B) Rubber is an insulator, so the loop is no longer completed electrically and no current can flow.
- (C) Rubber has a very low melting point, so it melted at once and broke the circuit like a fuse.
- (D) Rubber turns the cell's terminals around, reversing the current to zero.

13. Four materials are tested one at a time as the connecting link in a bulb circuit: a copper wire, a glass rod, a plastic ruler and a dry wooden pencil casing. With how many of them, and which, will the bulb glow?

- (A) Only with the copper wire, because copper is a conductor while glass, plastic and dry wood are insulators.
- (B) With all four, because anything placed in the gap completes the circuit.
- (C) With none, because a bulb needs at least two cells to light at all.
- (D) With the glass and plastic, because clear or smooth materials conduct best.

14. A pencil's graphite (the black 'lead') is placed in series in a bulb circuit and the bulb glows faintly. What does this best show?

- (A) The wooden casing of the pencil is what conducts, while the graphite is an insulator.
- (B) Graphite is acting as a fuse and is about to melt and break the circuit.
- (C) Graphite is a conductor of electricity, even though it is a non-metal, so it completes the loop.
- (D) The bulb is glowing because graphite generates its own current without a cell.

15. Two wires of the same material and same length are connected in turn across the same cell with the same bulb. With the thin wire the bulb is dim; with the thick wire it is bright. The best reason is that the thin wire:

- (A) Has a higher resistance than the thick wire, so it lets less current through and the bulb is dimmer.
- (B) Is a perfect insulator, which is why the bulb is only dim and not dark.
- (C) Is longer than the thick wire, which is why its resistance is higher.
- (D) Carries more current than the thick wire, which dims the bulb.

16. When an electric current flows through any ordinary wire, the wire warms up. A student claims a current can flow through a wire 'without producing any heat at all in an ordinary wire.' This claim is:

- (A) Correct, because heat appears only in special heater wires like nichrome.
- (B) Wrong; current passing through any ordinary resistive wire always produces some heating effect, however small.
- (C) Correct, because thick copper wires cool down when current flows.
- (D) Wrong, because current actually makes every wire glow with light.

17. A nichrome heating coil and a copper connecting wire are placed in the same series loop, so the same current passes through both. The nichrome glows red while the copper stays cool. The main reason is that:

- (A) The copper carries no current at all, so only the nichrome can heat up.
- (B) Nichrome has a much higher resistance than copper, so far more heat is produced in the nichrome for the same current.
- (C) Nichrome is magnetic and copper is not, and magnetism makes the nichrome hot.
- (D) Copper has a lower melting point than nichrome, so it never gets hot.

18. Two heating elements are made of the same nichrome wire of the same thickness, but element X is long and element Y is short. The same voltage is applied to each in turn.

Which gets the longer wire's behaviour right?

- (A) The longer wire X always gives more heat because there is simply more metal to warm up.
- (B) Length makes no difference to resistance, so both produce exactly the same heat.
- (C) The shorter wire Y has lower resistance, so a larger current flows and it produces more heat for the same voltage.
- (D) The longer wire X carries a larger current because current has more room to flow.

19. An electric room heater uses a nichrome coil rather than a copper coil. Which statement best explains nichrome's suitability?

- (A) Nichrome is a far better conductor than copper, so it heats faster.
- (B) Nichrome is magnetic, and the magnetism is what makes it glow red.
- (C) Nichrome has a very low melting point, so it softens gently and gives off heat.
- (D) Nichrome has high resistance and a high melting point, so it gets very hot and glows without melting away.

20. The filament of a glowing bulb is made of tungsten and is wound into a tight coil. Why a coiled tungsten filament rather than a straight copper one?

- (A) Tungsten has a very high melting point so it can become white-hot without melting, and coiling packs a long thin high-resistance wire into a small space.
- (B) Copper is used in filaments because it glows more brightly than tungsten.
- (C) The coil shape is only decorative and has no effect on how the bulb works.
- (D) Tungsten is chosen because it melts at a low temperature and so heats quickly.

21. Two identical immersion heaters are run, one for 1 minute and one for 2 minutes, in identical buckets of water at the same start temperature. Comparing the water afterwards:

- (A) Both buckets end at the same temperature, because the heater's power fixes a single final temperature.
- (B) The bucket heated for less time gets warmer, because a quick burst delivers more energy.

(C) The bucket heated for the longer time receives more heat, so its water ends up warmer.

(D) Neither bucket warms at all, because immersion heaters use the magnetic effect, not heat.

22. Which group lists only appliances that work mainly on the heating effect of electric current?

(A) Electric bell, crane electromagnet and loudspeaker.

(B) Ceiling fan, mixer-grinder and water pump.

(C) Electric iron, toaster and electric geyser.

(D) Magnetic compass, bar magnet and torch switch.

23. An electric fuse is a deliberately weak link in a circuit. The most precise description of how it protects the wiring is that it is a wire which:

(A) Has low melting point and gets hot fastest when current is excessive, so it melts and breaks the circuit before the main wiring overheats.

(B) Is a very good conductor with a high melting point, so it carries any current without trouble.

(C) Stores the extra electric charge safely until the current returns to normal.

(D) Speeds up the current so the appliances run more powerfully.

24. A circuit is meant to carry at most 5 A safely. Which fuse should be fitted, and why?

(A) A 50 A fuse, so it never melts and the supply is never interrupted.

(B) A 0.5 A fuse, so it melts even during ordinary safe use.

(C) No fuse at all, so that nothing can ever interrupt the supply.

(D) A 5 A fuse, so it carries the normal current but melts and cuts off the supply once the current rises above the safe 5 A limit.

25. After a fuse has blown, a person twists the two melted ends together with a length of thick copper wire so it 'will never blow again.' What is the real danger of this?

(A) The bulbs in the house will now glow much more dimly than before.

(B) The cells in the circuit will start charging in the wrong direction.

(C) The magnetic effect of the current will become dangerously strong.

(D) If the current becomes excessive again, nothing will melt to break the circuit, so the wiring can overheat and start a fire.

26. Several high-power appliances are switched on at one socket and the current grows very large until the fuse melts. The fuse melting in this case shows it is doing its job by responding to:

(A) A drop in current below normal, which fuses are designed to detect.

(B) An overload — too much current drawn at once — which it interrupts by melting before the wires can overheat.

- (C) The cells being inserted the wrong way round in the circuit.
- (D) The magnetic effect of the appliances pulling the fuse apart.

27. Why is a thin fuse wire designed to melt before the thicker household wiring overheats, when the same large current flows through both?

- (A) The thin fuse wire carries no current at all, so it is the only part that can melt.
- (B) The thin fuse wire is a much better conductor, so it gathers all the heat.
- (C) The thin fuse wire is magnetic, and magnetism makes it melt before the rest.
- (D) The thin fuse wire has higher resistance and a low melting point, so it heats up fastest and reaches its melting point first.

28. A safety leaflet states: 'A fuse protects a circuit by conducting the dangerous extra current safely to earth without ever breaking the circuit.' This statement is:

- (A) Correct; that is exactly how every fuse works.
- (B) Wrong; a fuse protects precisely by melting and breaking the circuit when the current is too high, not by carrying the excess current away.
- (C) Wrong, because a fuse actually increases the current to a safe level.
- (D) Correct, but only for the fuses used inside electric bells.

29. A modern protective device does the fuse's job but, after it cuts off the supply due to excess current, can simply be switched back on instead of being replaced. This device is the:

- (A) Tungsten filament of a bulb.
- (B) Electromagnet of an electric bell.
- (C) Electric cell of the circuit.
- (D) Miniature circuit breaker (MCB).

30. A homeowner keeps blowing the 5 A fuse, so they fit a 30 A fuse to 'stop the nuisance.' Why is this a poor idea even though the fuse stops blowing?

- (A) The 30 A fuse will melt too easily and keep cutting the supply more often.
- (B) The 30 A fuse lets a far larger, unsafe current flow before it melts, so the wiring can overheat dangerously without protection.
- (C) A higher-rated fuse reverses the current direction in the house wiring.
- (D) A 30 A fuse makes all the appliances run at half their power.

31. In 1820, Hans Christian Oersted saw a compass needle near a wire swing aside. The crucial condition that made the needle move was that:

- (A) An electric current was switched on in the wire, creating a magnetic field around it.
- (B) The wire was simply lying still with no current passing through it.
- (C) The wire had been painted a brighter colour.
- (D) The wire was bent into a shorter length.

32. A compass needle held below a straight wire deflects when current flows. The current is then switched off. What happens to the needle, and why?

- (A) It stays locked in the deflected direction forever, because the wire is now permanently magnetic.
- (B) It begins spinning round and round continuously without stopping.
- (C) It swings back to its original north–south direction, because the wire's magnetic field disappears when the current stops.
- (D) It melts, because switching off releases stored heat into the needle.

33. An engineer wants to check whether current is flowing in a sealed cable without cutting it open. Which observation would best confirm current is flowing?

- (A) The cable's outer colour changes when the supply is switched on.
- (B) A compass needle held close to the cable deflects when the supply is switched on.
- (C) The cable becomes noticeably heavier while the supply is on.
- (D) The cable becomes longer while the supply is on.

34. A current-carrying coil is wound around a soft-iron nail to form an electromagnet. Compared with the same length of straight wire carrying the same current, the coil-with-iron arrangement gives:

- (A) A much stronger magnetic effect, because coiling concentrates the field and the iron core greatly increases it.
- (B) No magnetic effect at all, because winding the wire cancels the magnetism.
- (C) A weaker magnetic effect, because the iron absorbs the magnetism.
- (D) Only a heating effect, because coiling removes the magnetic effect.

35. A student wants to make an electromagnet stronger but must keep the same single cell. Which single change is most sensible?

- (A) Remove the iron core from inside the coil.
- (B) Use fewer turns of wire on the core.
- (C) Wind many more turns of wire around the same iron core.
- (D) Replace the iron core with a wooden rod.

36. An electromagnet lifts at most 4 iron nails using one cell. The student then adds a second cell correctly in series and also adds extra turns of wire. The number of nails it can now lift will most likely:

- (A) Increase, because both more current and more turns make the electromagnet stronger.
- (B) Decrease, because two cells share the magnetic effect between them.
- (C) Stay exactly the same, because an electromagnet has a fixed strength.
- (D) Drop to zero, because adding turns cancels the magnetism.

37. A scrap-yard crane uses a large electromagnet over a mixed heap of iron rods, aluminium sheets, copper pipes and plastic crates. When switched on, the electromagnet will pick up:

- (A) Every object in the heap, whatever it is made of.
- (B) Only the plastic crates, because they are the lightest.

- (C) Nothing at all, because electromagnets cannot lift heavy objects.
- (D) The iron rods, but not the aluminium, copper or plastic objects.

38. Why is an electromagnet preferred over a permanent bar magnet for a crane that must lift and then release iron scrap?

- (A) An electromagnet works without any electric current, so it never costs anything to run.
- (B) An electromagnet is made of plastic, so it is lighter to lift with.
- (C) An electromagnet can be switched on to grip the load and switched off to drop it, whereas a permanent magnet cannot be turned off.
- (D) An electromagnet is always magnetic and so can never accidentally drop a load.

39. In an electric bell, when current flows, the coil on the iron core becomes a magnet and pulls the iron armature so the hammer strikes the gong. At that instant a contact is broken. What happens next, making the bell ring repeatedly?

- (A) The current grows larger each time until the bell finally melts.
- (B) The electromagnet turns into a permanent magnet and holds the armature still.
- (C) The current stops, the electromagnet loses its magnetism, a spring pulls the armature back, the contact remakes, and the cycle repeats over and over.
- (D) The bell strikes the gong only once and then stays silent for good.

40. A device or material is needed that becomes magnetic only while current flows and loses its magnetism instantly when the current stops. Which fits this description?

- (A) A permanent steel bar magnet.
- (B) A copper connecting wire on its own.
- (C) A soft-iron core wound with a current-carrying coil (an electromagnet).
- (D) A tungsten bulb filament.

41. Which single statement correctly describes the relationship between the heating effect and the magnetic effect of a current in a wire?

- (A) A current produces only a heating effect and never a magnetic one.
- (B) A current in a wire produces both a heating effect and a magnetic effect at the same time.
- (C) A current produces only a magnetic effect and never a heating one.
- (D) Heat and magnetism appear only after the current has been switched off.

42. Two coils are wound from the same wire with the same number of turns and carry the same current. Coil P has a soft-iron core inside; coil Q has only air inside. Which is the stronger magnet, and why?

- (A) Coil Q, because air lets the magnetism spread out more freely.
- (B) They are exactly equal, because the core makes no difference to coil strength.
- (C) Neither, because a coil with the same current in both cannot be magnetic.
- (D) Coil P, because the soft-iron core greatly increases the magnetic effect of the coil.

43. A learner says: 'Once I switch on an electromagnet and lift some nails, the nails will keep clinging even after I switch the current off.' Why is this generally wrong for a soft-iron electromagnet?

- (A) When the current stops, the soft-iron core loses almost all its magnetism, so the nails fall off.
- (B) The nails keep clinging because soft iron stays a strong magnet for ever.
- (C) The nails fall only if the coil is also unwound from the core.
- (D) The current never really switches off in an electromagnet.

44. An open switch in a working bulb circuit is then closed. Tracing the change, the bulb lights because closing the switch:

- (A) Bridges the gap and completes the loop, so current can now flow all the way round through the bulb.
- (B) Adds an extra cell to the circuit, raising the voltage.
- (C) Increases the bulb's resistance until it glows.
- (D) Reverses the direction of the cell's terminals.

45. Two cells are connected positive-to-positive and negative-to-negative (opposing each other) and placed in a loop with a good bulb and closed switch. The bulb stays dark. The best reason is that:

- (A) The two cells push current in opposite directions, so their voltages cancel and almost no net current flows.
- (B) Opposing cells double the voltage, which instantly destroys the bulb.
- (C) Opposing cells form a short circuit that bypasses the bulb.
- (D) The switch must be the cause, since two cells always light a bulb.

46. A bulb circuit with one cell glows. The single cell is replaced by a battery of two cells in series the correct way (+ to -). A child predicts the bulb will get dimmer. The child is:

- (A) Right, because two cells must share the work and so each pushes less.
- (B) Wrong; adding a cell correctly in series increases the push and makes the bulb glow brighter.
- (C) Right, but only for bulbs with tungsten filaments.
- (D) Wrong, because the bulb will instead go out completely.

47. Which pairing of a circuit component with its correct job is right?

- (A) Bulb — to break the circuit automatically when the current is too high.
- (B) Switch — to make or break the circuit so current can be turned on or off.
- (C) Fuse — to store electric charge for later use.
- (D) Cell — to measure how much current is flowing.

48. A toaster and a ceiling fan are running from the same supply. Which effect of current does each mainly use?

- (A) The toaster uses the magnetic effect, while the fan uses the heating effect.
- (B) Both the toaster and the fan use the heating effect.
- (C) Both the toaster and the fan use the magnetic effect.
- (D) The toaster mainly uses the heating effect, while the fan mainly uses the magnetic effect (in its motor).

49. An immersion water heater and an electric bell are compared. They rely, respectively, on which effects of electric current?

- (A) The immersion heater uses the heating effect; the electric bell uses the magnetic effect.
- (B) The immersion heater uses the magnetic effect; the bell uses the heating effect.
- (C) Both use the heating effect.
- (D) Both use the magnetic effect.

50. Identify the statement that is INCORRECT about electromagnets.

- (A) An electromagnet can be switched on and off, while a permanent bar magnet cannot.
- (B) An electromagnet becomes stronger when more turns are added or more current flows.
- (C) An electromagnet attracts iron and steel objects when current flows.
- (D) An electromagnet attracts plastic, aluminium and wood just as strongly as it attracts iron.

51. A loop is drawn with a cell and a bulb in series, plus a separate plain wire connecting the cell's two terminals directly to each other. With everything connected, the bulb most likely:

- (A) Glows brighter, because the extra wire delivers more current to it.
- (B) Glows normally, because an extra wire across the cell changes nothing.
- (C) Stays dim or dark, because the direct wire offers a low-resistance shortcut that carries most of the current around the bulb.
- (D) Cannot be connected at all, because a cell can have only one wire.

52. A bulb circuit has TWO switches in series in the same single loop. For the bulb to glow, the switches must be:

- (A) Either one closed is enough, because one closed switch can complete the loop alone.
- (B) Both open, so that the current is free to flow.
- (C) One open and one closed, to balance the circuit.
- (D) Both closed, because either one being open leaves a gap that breaks the single loop.

53. A bulb circuit has TWO bulbs in series in one loop, and one of the bulbs has a broken (fused) filament. What happens to the other bulb?

- (A) The other bulb glows twice as bright, because all the current now goes through it.
- (B) The other bulb glows normally, because each bulb has its own separate path.

- (C) The other bulb also goes dark, because the broken filament breaks the single loop and stops the current everywhere.
- (D) The other bulb explodes from the extra current.

54. A diagram shows a battery symbol (several cell symbols in a row), then a closed switch, then a bulb, then back to the battery. A student reads it as 'open circuit, bulb off.' Why is the student wrong?

- (A) A battery of several cells always leaves the circuit open.
- (B) The switch shown is closed and the loop is unbroken, so it is a closed circuit and the bulb should glow.
- (C) A bulb in a loop always means the circuit is open.
- (D) Any circuit drawn on paper is automatically an open circuit.

55. A bulb glows normally on a fresh cell. Over many hours of use it slowly dims and goes out, while all wires and the switch test fine. The most likely reason is that:

- (A) The connecting wires have turned into insulators over time.
- (B) The closed switch has slowly opened the circuit by itself.
- (C) The cell has run down, so it no longer provides enough push to drive a useful current.
- (D) The bulb's glass has thickened and is blocking the light.

56. A heater element and a torch bulb both rely on the heating effect, yet only the bulb gives bright light. The key difference is that the bulb's filament:

- (A) Uses the magnetic effect while the heater uses the heating effect.
- (B) Has lower resistance than the heater, so it stays cool and still shines.
- (C) Is made of copper, which is why it lights up.
- (D) Reaches a much higher (white-hot) temperature than the heater element, so it glows brightly with visible light.

57. A coil of insulated copper wire around an iron nail is connected to a cell through a switch, forming an electromagnet that holds some pins. The switch is then opened. The pins fall because:

- (A) Opening the switch makes the iron core a stronger magnet that flings the pins off.
- (B) Opening the switch heats the coil, melting the pins away.
- (C) The pins fall only because gravity suddenly increased.
- (D) Opening the switch stops the current, so the coil's magnetic field collapses and the iron core loses its magnetism.

58. A factory uses a large electromagnet on a conveyor to separate a mixture. Which mixture can it usefully separate by attracting one part?

- (A) Iron filings mixed with sand, by attracting the iron and leaving the sand behind.
- (B) Salt mixed with sugar, by attracting the salt grains.

- (C) Sand mixed with sawdust, by attracting the sand.
- (D) Copper turnings mixed with aluminium, by attracting the copper.

59. A bell's springy contact strip is bent so that it no longer touches the contact screw even when the armature is at rest. With current switched on, the bell will:

- (A) Not ring at all, because the broken contact means the circuit never closes, so no current reaches the electromagnet.
- (B) Ring continuously and louder than before, because the gap speeds up the cycle.
- (C) Ring exactly once and then stop, because the electromagnet fires only once.
- (D) Heat up and melt, because the open contact forces a huge current.

60. A student claims: 'A short, thick copper wire makes a great fuse because it can carry huge currents without melting.' Why is this exactly backwards for fuse design?

- (A) A good fuse must be a thin, low-melting wire that melts under excess current; a thick high-melting copper wire would never melt and so would never protect the circuit.
- (B) It is correct, since a fuse should carry as much current as possible without melting.
- (C) A copper fuse would melt too easily, cutting off the supply during normal use.
- (D) Copper cannot be used in any electrical circuit, so it would block all current.

Solutions — Science (Class 7), Chapter 14:

Electric Current and Its Effect — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	D	21	C	31	A	41	B	51	C
2	C	12	B	22	C	32	C	42	D	52	D
3	B	13	A	23	A	33	B	43	A	53	C
4	B	14	C	24	D	34	A	44	A	54	B
5	A	15	A	25	D	35	C	45	A	55	C
6	C	16	B	26	B	36	A	46	B	56	D
7	D	17	B	27	D	37	D	47	B	57	D
8	B	18	C	28	B	38	C	48	D	58	A
9	D	19	D	29	D	39	C	49	A	59	A
10	C	20	A	30	B	40	C	50	D	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Standard convention: long thin line = positive terminal, short thick line = negative terminal; the student called the short thick line positive, so it is reversed. The lines are terminals of the cell, not wires (wires are plain lines), so the 'connecting wires' option fails. Thickness is purely a drawing convention for marking the negative side, not a measure of cell strength. The symbol's meaning does not depend on whether it is drawn left or right, so position cannot decide the sign.

2. (C) Voltages add only when cells are in series, i.e. + of one to – of the next, giving $2+2+2 = 6$ V. Joining all positives together and all negatives together is a parallel arrangement that keeps the voltage at one cell (2 V), not 6 V. Joining + to + and – to – makes cells oppose, so their pushes partly cancel rather than add. Air gaps break the conducting path entirely, so no current and no battery action.

3. (B) When cells oppose (+ to +), their voltages subtract; two equal cells leave roughly zero net push, so the bulb stays dark — that is the correct reason. The cells are not

individually too weak; the issue is cancellation, not weakness. Opposing cells do not double the voltage — that would require series aiding (+ to –), so the 'burns out' option is wrong. Two correctly arranged cells routinely light torch bulbs, so 'a torch must use one cell' is false.

4. (B) Series-aiding cells add their pushes, so each extra cell raises the voltage and current, making the bulb brighter — and excess current can burn the filament, which is the full correct statement. Series cells do not 'share' current; current is the same all round the loop, so the dimmer option is wrong. Brightness is set by the total voltage, not capped by the first cell, so 'stays the same' fails. The number of cells being odd or even has nothing to do with completing a loop.

5. (A) Conventional current leaves the positive (long thin) terminal, travels the external loop, and returns to the negative (short thick) terminal — the correct path. Starting at the negative terminal reverses this, so that option is wrong. Current does not emerge from the 'middle' of a cell in two directions; it follows the single loop. Current must complete the whole loop back to the cell, so 'never back to the cell' is impossible for a working circuit.

6. (C) A small circle with a cross is the standard symbol for a bulb; with a cell and a closed switch the loop is complete, so the bulb glows — the correct reading. A cell is drawn as long-thin and short-thick lines, not a crossed circle, so 'cell' is wrong. An open switch is a raised arm leaving a gap, not a crossed circle. A fuse is usually shown as a thin line in a rectangle, not a crossed circle, so the fuse reading is also wrong.

7. (D) An arm lying flat across both contacts bridges the gap, making the switch closed and the loop complete, so the bulb can glow — correct. An open switch is the raised-arm form leaving a gap, not the flat form, so 'open' is wrong. A switch arm lying flat is its normal closed position, not a melted fuse. A switch arm has nothing to do with recharging a cell, so that option is irrelevant.

8. (B) Having a cell and bulb is not enough; the loop must be unbroken, and the open switch breaks it, so no current flows — that is the correct reason. A cell does not need 'charging' to drive a bulb, so that option is false. Connecting wires have very low resistance and do not 'absorb' the current, so the bulb-receives-nothing claim is wrong. A single fresh cell can easily light a torch bulb, so 'never enough push' is false.

9. (D) A low-resistance wire across the bulb gives current an easy bypass, so most current avoids the high-resistance filament and the bulb stays dark — correct. The extra path steals current from the bulb rather than feeding it more, so 'brighter' is wrong. The short clearly does change behaviour, so 'nothing changes' is false. The cell is heavily loaded but does not become permanently dead just from a brief short, so 'stops forever' overstates it.

10. (C) A break anywhere — even hidden inside insulation — makes the circuit open, so no current flows and the bulb stays dark; that is correct. A circuit with a break is not closed, so

'closed circuit, wrong colour' is wrong (and bulb colour is irrelevant to glowing). A short circuit is an unintended low-resistance path, not a break, and would not be caused by a snapped wire. A single cell does not turn into a two-cell battery, so that option is nonsense.

11. (D) Dry wood is an insulator; placed in series it breaks the conducting path, so no current flows and the bulb stays dark — correct. Wood is a poor conductor, not a good one, so the 'carries current away' option is the opposite of the truth. Wood is not a metal and does not melt like a fuse wire. Magnetism does not block ordinary current flow, and wood is not magnetic, so that option fails.

12. (B) Rubber is a classic insulator; using it in place of a wire leaves the loop electrically incomplete, so no current flows and the bulb is dark — correct. Rubber conducts far worse than metal, not better, so the 'better conductor' option is wrong. The bulb fails immediately on connection, not because rubber melted; rubber is not a fuse element. Inserting rubber does not flip the cell's terminals, so the 'reverses current' option is false.

13. (A) Of the four, only copper is a metal conductor; glass, plastic and dry wood are insulators that block the loop, so the bulb glows only with copper — correct. Not 'anything' completes a circuit; insulators do not, so 'all four' is wrong. A single cell can light a torch bulb, so 'needs two cells' is false. Being clear or smooth has nothing to do with conduction; glass and plastic are insulators, so that option fails.

14. (C) Graphite conducts electricity despite being a non-metal, so it completes the loop and the bulb glows — the correct lesson. The wooden casing is an insulator; if it carried the current the bulb would not glow, so that option is wrong. A faint steady glow does not mean the graphite is melting like a fuse. Graphite produces no current of its own; the cell drives the current, so the 'generates its own current' option is false.

15. (A) For the same material and length, a thinner wire has higher resistance, allowing less current, so the bulb is dimmer — correct. The wire is not an insulator (the bulb still glows dimly), so 'perfect insulator' is wrong. The two wires are stated to be the same length, so the higher resistance is due to thinness, not length. More current would make the bulb brighter, not dimmer, so the 'carries more current' option is self-contradictory.

16. (B) Every ordinary wire has some resistance, so current always produces at least a little heat — the claim is wrong, which is the correct verdict. Heating is not unique to nichrome; nichrome simply heats much more, so 'only special wires' is false. Carrying current never cools a wire below its surroundings, so the 'copper cools down' option is wrong. Most wires get warm without glowing; only very hot filaments emit visible light, so 'every wire glows' is false.

17. (B) Same current through both means the higher-resistance nichrome produces much more heat, so it glows while copper stays cool — correct. In a series loop the same current flows through both, so 'copper carries no current' is false. Heating is due to resistance, not

magnetism, and being magnetic would not heat a wire, so that option is wrong. Copper actually has a lower melting point than nichrome, and melting point does not stop a wire heating; this reasoning is backwards.

18. (C) For the same voltage, the shorter wire has lower resistance and so passes more current, producing more heat — correct. 'More metal' does not guarantee more heat; the long wire's higher resistance limits the current, so X does not automatically win. Length does change resistance (longer = more resistance), so 'no difference' is false. A longer wire has higher resistance and so carries less current, not more, making the 'more room' option wrong.

19. (D) Nichrome's high resistance produces strong heating and its high melting point lets it glow red without melting — exactly why it is chosen. A better conductor (like copper) has lower resistance and heats less, so 'far better conductor' is the opposite of what is wanted. Glowing comes from heat, not magnetism, so the magnetic option is wrong. A low melting point would make the element melt when hot; nichrome's melting point is high, so that option is false.

20. (A) Tungsten's extremely high melting point lets the filament reach white heat without melting, and the coil fits a long, thin, high-resistance wire into a tiny space — the correct full reason. Copper would melt long before reaching glowing temperatures, so 'copper glows more brightly' is false. The coil is functional, not decorative, so that option is wrong. A low melting point would be a fatal flaw for a filament; tungsten's point is high, so that reasoning is backwards.

21. (C) More running time means more heat delivered, so the longer-heated water is warmer — correct. There is no fixed 'final temperature' from power alone; longer heating keeps adding energy, so 'same temperature' is wrong. A shorter time delivers less energy, not more, so the 'quick burst' option is false. Immersion heaters work on the heating effect, not the magnetic effect, so the last option misidentifies the principle.

22. (C) An iron, a toaster and a geyser all heat a high-resistance element, so they rely on the heating effect — correct. A bell, crane magnet and loudspeaker rely on the magnetic effect, not heating. A fan, mixer and pump use motors (magnetic effect) to produce motion, not heat. A compass and bar magnet are not even powered appliances, and a torch switch only opens or closes a circuit, so that set is wrong.

23. (A) A fuse is a thin, low-melting-point wire that overheats first under excess current and melts to open the circuit, protecting the rest — correct. A wire that carries 'any current' would never melt and so could not protect anything, so the high-melting option is wrong. A fuse does not store charge; it simply melts when overloaded, so the 'stores charge' option is false. A fuse does not speed up current; that misunderstands its purpose entirely.

24. (D) A fuse rated at the safe limit (5 A) passes normal current but blows just above it, giving protection — correct. A 50 A fuse would let dangerously large currents flow without melting, defeating the protection, so it is wrong. A 0.5 A fuse would blow constantly during normal use, making the circuit unusable. Fitting no fuse removes all protection, which is unsafe, so that option fails.

25. (D) A thick copper link never melts, so an overload is no longer cut off and the wiring can overheat and catch fire — the correct danger. The fix does not dim bulbs; it removes protection, so the 'dimmer bulbs' option misses the point. Mains wiring has no cells to charge backwards, so that option is irrelevant. The hazard is overheating/fire, not a strong magnetic effect, so the last option is wrong.

26. (B) Drawing too much current at once is an overload, and the fuse melts to interrupt it — correct. Fuses respond to excessive current, not to a current that is too low, so the 'drop in current' option is wrong. Mains appliances are not cell-powered, so 'cells the wrong way' is irrelevant. The fuse melts from heating, not from a magnetic pull, so the magnetic option is false.

27. (D) Being thin (higher resistance) and made of low-melting metal, the fuse wire heats fastest and melts first, protecting the thicker wiring — correct. In series the same current flows through the fuse and the wiring, so 'carries no current' is false. A better conductor has lower resistance and heats less, so 'better conductor' is the opposite of the truth. Melting is caused by heating, not magnetism, so the magnetic option is wrong.

28. (B) A fuse works by melting to open the circuit, not by quietly conducting the surplus current onward — so the statement is wrong, which is the correct verdict. It is not 'exactly how every fuse works,' since fuses break rather than conduct excess, so that option is false. A fuse does not raise the current; it cuts it off, so the 'increases current' option is wrong. Electric bells do not contain protective fuses of this kind, so that qualifier is meaningless.

29. (D) An MCB trips on overload and is reset by flicking it back on, replacing the one-use fuse — correct. A tungsten filament is the light-emitting part of a bulb and offers no resettable protection. An electromagnet is a coil-and-core magnet, not a circuit protector. A cell supplies energy and does not protect against overload, so it is wrong.

30. (B) A 30 A fuse tolerates much larger currents before melting, removing protection and risking overheated wiring — the correct concern. A higher rating melts less easily, not more, so the 'cuts supply more often' option is backwards. Fuse rating does not reverse current direction, so that option is nonsense. A bigger fuse does not halve appliance power; it just permits more current, so the last option is wrong.

31. (A) The needle deflected only when current flowed, because a current produces a magnetic field — that is Oersted's discovery and the correct condition. A wire with no current produces no field, so the still-wire option cannot deflect the needle. Painting a wire

changes nothing electrically or magnetically. Merely shortening the wire does not start a current, so that option is also wrong.

32. (C) The wire's magnetic field exists only while current flows, so when it stops the needle returns to north–south — correct. The wire is not permanently magnetised by a brief current, so the 'locked forever' option is wrong. Switching off does not make the needle spin endlessly; it settles back, so that option is false. There is no stored heat that melts a compass needle on switch-off, so the 'melts' option is absurd.

33. (B) A current produces a magnetic field, so a nearby compass deflecting on switch-on confirms current flows without opening the cable — correct. Current does not change a cable's colour, so that observation proves nothing. Carrying current does not add mass, so 'heavier' is false. A cable does not measurably lengthen because current flows, so the 'longer' option is wrong.

34. (A) Winding the wire into a coil adds the fields of many turns, and a soft-iron core multiplies the effect, so the electromagnet is much stronger — correct. Coiling reinforces rather than cancels the field, so 'no effect' is wrong. The iron does not absorb the magnetism; it strengthens it, so 'weaker' is false. A coil still has both effects; coiling does not remove magnetism, so the 'only heating' option is wrong.

35. (C) With the cell fixed, adding more turns increases the magnetic effect, so it is the sensible choice — correct. Removing the iron core weakens the magnet, the opposite of the goal. Fewer turns also weaken it, so that option is wrong. A wooden rod is non-magnetic and does not strengthen the field the way iron does, so replacing iron with wood weakens the electromagnet.

36. (A) More current (two cells in series) and more turns both strengthen the electromagnet, so it lifts more nails — correct. Cells in series add their push and do not 'share' the effect, so 'decrease' is wrong. An electromagnet's strength is adjustable, not fixed, so 'stay the same' is false. Extra turns add to, rather than cancel, the magnetism, so the 'drop to zero' option is wrong.

37. (D) An electromagnet attracts magnetic materials such as iron and steel, so it lifts the iron rods but leaves aluminium, copper and plastic — correct. It does not attract every material, so 'every object' is wrong. Plastic is non-magnetic and is not picked up at all, so 'only the plastic crates' is false. Electromagnets in cranes do lift heavy iron loads, so 'nothing at all' is wrong.

38. (C) An electromagnet's magnetism can be switched on and off with the current, letting the crane grab and release scrap — the correct advantage. It does need current to work, so the 'works without current' option is false. Electromagnets are built from coil and iron core, not plastic, so that option is wrong. The very point is that it can be switched off to drop the load, so 'always magnetic, never drops' contradicts its useful behaviour.

39. (C) Breaking the contact stops the current, the electromagnet de-magnetises, the spring returns the armature, the contact remakes and the whole cycle repeats — giving continuous ringing, which is correct. The current does not grow each cycle, and the bell does not melt, so that option is wrong. The core is soft iron, not a permanent magnet, so it cannot hold the armature, making that option false. Because the cycle repeats, the bell does not stop after one strike, so the 'once and silent' option is wrong.

40. (C) An electromagnet (soft-iron core plus coil) is magnetic only while current flows and loses it when the current stops — exactly the described behaviour. A permanent steel magnet keeps its magnetism with no current, so it does not fit. A plain copper wire carries current but is non-magnetic as a material and makes a very weak field without a core or coil, so it is not the intended answer. A tungsten filament heats and glows; it is not a magnet, so it is wrong.

41. (B) A current simultaneously warms the wire (heating effect) and sets up a magnetic field around it (magnetic effect) — both occur together, which is correct. It is not heating only, since the magnetic effect is real (Oersted), so that option is wrong. It is not magnetic only, since resistive heating always occurs, so that option is wrong. Both effects exist while current flows, not after switch-off, so the last option is false.

42. (D) A soft-iron core multiplies a coil's magnetic effect, so the iron-cored coil P is much stronger — correct. Air does not boost magnetism the way iron does, so the air-cored Q is weaker, not stronger. The core makes a large difference, so 'exactly equal' is false. A current-carrying coil is magnetic whether or not it has a core, so 'neither is magnetic' is wrong.

43. (A) A soft-iron electromagnet loses its magnetism when the current stops, so the nails drop — that is the whole point and the correct reason. Soft iron does not stay strongly magnetised after switch-off, so 'stays a magnet for ever' is wrong. The nails fall as soon as the current stops, without any need to unwind the coil, so that option is false. The current does switch off when the circuit is broken, so 'never switches off' is incorrect.

44. (A) Closing the switch removes the gap, completing the loop so current flows through the bulb and it lights — correct. A switch does not add a cell or raise the voltage, so that option is wrong. Closing a switch does not increase the bulb's resistance; the bulb glows because current can now flow, so that option is false. A switch does not flip the cell's terminals, so the 'reverses terminals' option is wrong.

45. (A) Cells facing against each other have voltages that subtract, leaving nearly zero net push, so the bulb stays dark — correct. Opposing cells subtract rather than double the voltage, so the 'double, destroys bulb' option is wrong. Opposing cells do not create a low-resistance bypass; the dark bulb is due to cancellation, not a short. The switch is closed and good; the real cause is the opposing cells, so 'must be the switch' is false.

46. (B) Series-aiding cells add their voltages, so the current and brightness increase — the child's 'dimmer' prediction is wrong, which is the correct verdict. Cells in series do not split the work; current is the same throughout, so the 'share the work' reasoning is false. The result does not depend on the filament being tungsten, so that qualifier is meaningless. The bulb brightens rather than going out, so the 'goes out' option is also wrong.

47. (B) A switch makes or breaks the loop to turn current on or off — correct. Breaking the circuit on excess current is the fuse's job, not the bulb's, so that pairing is wrong. A fuse melts to break the circuit; it does not store charge, so 'fuse — store charge' is false. A cell supplies energy; it does not measure current, so the last pairing is wrong.

48. (D) A toaster heats a high-resistance element (heating effect) and a fan spins a motor driven by magnetism (magnetic effect) — correct. Swapping the two is wrong, since toasters do not spin and fans do not heat to cook, so the swapped option fails. A fan does not rely on heating, so 'both heating' is wrong. A toaster's job is heat, not motion, so 'both magnetic' is also wrong.

49. (A) An immersion heater warms water through a resistive element (heating effect) and a bell uses an electromagnet (magnetic effect) — correct. The two are not swapped: a heater does not work by magnetism nor a bell by heating, so the swapped option is wrong. A bell does not heat water, so 'both heating' fails. A heater does not ring by magnetism, so 'both magnetic' is also wrong.

50. (D) Electromagnets attract magnetic materials like iron and steel, but not plastic, aluminium or wood, so the claim that it attracts them 'just as strongly as iron' is the incorrect statement — the right pick. The on/off control over a permanent magnet is a true property. Strength rising with more turns or current is true. Attracting iron and steel while current flows is true. So the three true statements are not the answer.

51. (C) A plain wire straight across the cell's terminals is a low-resistance short that draws most of the current away from the bulb, so the bulb is dim or dark — correct. The short steals current rather than giving the bulb more, so 'brighter' is wrong. A direct short across the cell clearly changes the behaviour, so 'changes nothing' is false. A cell can have many wires connected; nothing forbids two, so the 'only one wire' option is wrong.

52. (D) In a single series loop, both switches must be closed; an open switch anywhere breaks the loop and stops current — correct. With switches in series, closing only one still leaves the other gap open, so 'either one is enough' is wrong. Open switches break the loop, so 'both open' would give no current. Mixing one open and one closed still leaves an open gap, so the 'balance' option is false.

53. (C) In a series loop a broken filament opens the whole loop, so no current flows and the second bulb is also dark — correct. With the loop broken, no current flows at all, so the second bulb cannot glow brighter, ruling out 'twice as bright.' Series bulbs share one path,

not separate paths, so 'glows normally' is false. With zero current there is nothing to make a bulb explode, so that option is wrong.

54. (B) A closed switch in an unbroken loop makes a closed circuit, so the bulb should glow — the student misread it. A multi-cell battery does not force the circuit open; it just supplies more voltage, so that option is false. Having a bulb in the loop does not make the circuit open, so that reasoning is wrong. A circuit being on paper does not make it open; the switch state decides, so the last option is false.

55. (C) A cell gradually discharges with use, weakening the current until the bulb goes out — correct, especially as wires and switch test fine. Copper wires do not transform into insulators with use, so that option is wrong. A closed switch does not open itself, and the switch tested fine, so that option is false. Bulb glass does not thicken to block light; the dimming is from the failing cell, so the last option is wrong.

56. (D) The very thin tungsten filament reaches white heat and emits bright light, while the heater element glows only dull red — correct. Both work by the heating effect; the bulb does not use the magnetic effect, so that option is wrong. The filament does not stay cool; it must get extremely hot to glow, so 'lower resistance, stays cool' is false. Filaments are tungsten, not copper, and copper would melt before glowing, so the last option is wrong.

57. (D) An open switch cuts the current, the field collapses, and the soft-iron core loses its magnetism, so the pins fall — correct. Stopping the current weakens the magnet to nearly nothing rather than strengthening it, so 'stronger magnet flings them' is wrong. No current means no heating, so 'melts the pins' is false. Gravity is constant; the pins fall because the magnetism vanished, so the 'gravity increased' option is absurd.

58. (A) An electromagnet attracts magnetic iron, so it pulls iron filings out of a sand mixture, leaving the non-magnetic sand — correct. Salt and sugar are both non-magnetic, so the magnet cannot pick out the salt, ruling that out. Sand and sawdust are both non-magnetic, so the magnet attracts neither. Copper and aluminium are both non-magnetic, so the magnet cannot separate them either.

59. (A) If the springy strip never touches the screw, the circuit cannot close, so no current flows to the electromagnet and the bell stays silent — correct. A permanent gap means the loop never closes, so it cannot ring louder or faster, ruling out 'continuous and louder.' With no current ever flowing, even a single strike is impossible, so 'rings once' is wrong. An open contact carries no current, so there is nothing to overheat or melt, making the last option false.

60. (A) A fuse protects by melting when the current is too high, so it must be thin and low-melting; a thick high-melting copper wire would never melt and would give no protection — the correct explanation. The claim is not correct, because a fuse that never melts cannot protect, so the 'it is correct' option is wrong. Thick high-melting copper melts

less easily, not more, so 'melts too easily' is backwards. Copper is widely used for connecting wires, so 'cannot be used in any circuit' is false.